

HOLY FAMILY NURSING AND MIDWIFERY TRAINING COLLEGE

BEREKUM

A PATIENT/FAMILY CARE STUDY ON

HYPERTENSION

SERWAA DORECIA GYAN

4120220175

**A PATIENT/FAMILY CARE STUDY SUBMITTED TO THE NURSING AND
MIDWIFERY COUNCIL OF GHANA IN PARTIAL FULFILMENT TOWARDS THE
AWARD OF LICENSE TO PRACTICE AS A PROFESSIONAL REGISTERED NURSE.**

AUGUST, 2024

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PREFACE

The patient and family care study is a detailed written account or report of the comprehensive individualized nursing care rendered to a particular patient and family within a specific period of time. This includes the study of the diagnosis, treatment and the actual nursing care rendered to the patient to meet his/her physical, psychological, social and spiritual needs. It involves the interaction between the patient, his/her family, the community in which he/she stays and the health team.

It forms part of the final assessment of the student nurse at the end of the three-year training program for a license to practice as a Registered General Nurse awarded by the Nursing and Midwifery Council of Ghana.

It presents the student the opportunity to put into practice the knowledge he or she acquires during training to give effective nursing care to a patient with reference to the patient's condition. The care also helps the student nurse to acquire more knowledge about the signs and symptoms, diagnosis, causes and treatment of a specific disease condition managed by the student.

The study serves as a reference paper for other student nurses and qualified health personnel who may be interested in its content. The patient's initials were used to maintain confidentiality.

ACKNOWLEDGMENT

I am mostly indebted to God almighty in heaven for all that he showered on me for the successful completion of this work; may His name be praised now and forever. I am grateful to my client, Mr. A. K. B. and his family members for their incredible co-operation. I am much thankful to them for all the information they provided towards the progress of this study. The next big ups also go to my supervisor Mr. Owusu Amos and the entire tutors of Holy Family Nursing and Midwifery Training College- Berekum, for their guidance, time and support in the course of writing this care study. I do say that God bless them so much. My sincere appreciation goes to my parents, Mr. S. G. and Madam A. A. and my siblings for their numerous support both in cash and kind during the period of this study. May the father in heaven replenish whatever they have given out to make this study a success. I would also like to express my gratitude to all nurses and doctors of the Holy Family Hospital- Berekum especially all the staff of the Males ward that helped me in diverse ways to make this study a success; I say thank you. I do not leave out authors and publishers whose works were used as literature in this study. My profound gratitude to them all.

Finally, I am grateful to all my course mates for their support during the writing of this care study.

INTRODUCTION

The care study is the sum total of the nursing care rendered to a patient and his/her family to attain and maintain a high level of wellness of the patient. It involves an interaction which could be as brief as a day or as long as the patient care last or to a peaceful death. This care study was made on Mr A. K. B., a 76 year old man who was admitted on 19th August, 2023 at the males' ward of Holy Family Hospital- Berekum with the diagnosis of Hypertension and discharged on 23rd August, 2023.

Reassurance, health education and follow up were made during the period of hospitalization and after discharge to ensure continuity of care. Patient was fully conscious at the time of admission.

The script is organized into six chapters;

Chapter one includes; assessment of patient/family, patient's particulars, family medical history, family socioeconomic history, patient's developmental history, patient's lifestyle/hobbies, past medical history, present medical history, admission of client, patient's concept of illness, literature review and validation of data.

Chapter two entails; analysis of data, comparison of data with standards, diagnostic investigation/tests, causes of patient's condition, clinical features, treatment, pharmacology of drugs, complications, client's /family strength, client health problems and nursing diagnosis.

Chapter three is made up of; planning for patient/family care, objectives/outcome criteria and nursing care plan.

Chapter four gives an account of; implementation of client/family care plan, summary of actual nursing care, preparation of client /family for discharge and rehabilitation and follow up.

Chapter five covers; the evaluation of care rendered to client/family, statement of evaluation, amendment of nursing care plan for partially met and unmet outcome criteria, termination of care

Chapter six encompasses; summary and conclusion as well as bibliography.

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CHAPTER ONE

ASSESSMENT OF THE PATIENT/FAMILY

1.0 Introduction

To provide holistic care to the patient and family, the nursing process must be employed.

Assessment is the first step/phase of the nursing process and it is an integral part of the care study. Assessment is the critical analysis and evaluation or judgment of the status or quality of a particular condition, situation or other subjects of appraisal (Weller, 2019). The purpose of assessment is to establish a data base concerning a client's physical, psychosocial and emotional health to identify health promoting behavior as well as actual and/or potential problems. The goal of assessment is to collect and analyze data to be used in formulating nursing diagnoses, identifying outcome, planning care, and developing nursing interventions. It commences from the day of admission until the care is terminated and this phase comprises concrete of data gathered systematically and tactfully. Information was gathered through face to face interaction with patient, critical observation of patient and various physical examination and laboratory investigations. The data collected would help the nurse to decide with other health professionals, patient and family on the suitable method to employ to attain optimal health. Data gathered helps to evaluate the causes or predisposing factors of the client's condition.

1.1 Patient's particulars

Particulars are details or information about a person or an event, especially when officially recorded (McIntosh, 2013).

Mr. A. K. B. is a 76-year-old male who was born on 20th March, 1947 in Biadan, in the Berekum municipality, in the Bono Region. He was born to Mr. K. A. and Mrs. A. S. who are all of blessed memory. He is dark in complexion and the first child among the five children of his parents.

Three of these are males. He is married with six children, of equal sex division. The children are

all alive and into various professions and businesses. Currently he lives at Senase, also in the Berekum municipality. He lives in a four-bed room apartment with the house number SE/109. His next of kin is his daughter Mrs. N. Y. Y. who is a hairdresser. He is a farmer. Mr. A. K. B. is a devoted Christian who worships at Methodist church of Ghana, Senase. He is a Ghanaian and an Akan by tribe, who speaks only two language. He is 1.7 metres tall and weighs 65 kilograms. He has no physical impairment.

1.2 Family medical history

Health history is a series of questions used to provide an overview of the patient's current health status. Attention is focused on the impact of psychosocial, ethnic, and cultural background on a person's health. Information is obtained on both paternal and maternal sides of family (Hinkle, Cheever & Overbaugh, 2022).

According to Mr. A. K. B., he has been admitted to the hospital quite severally on accounts of elevated blood pressure. Patient further added the family occasionally experiences minor ailments such as headaches, fever and body pains which are usually treated with over-the-counter medications like diclofenac, aspirin, paracetamol and sometimes reports to the nearest clinic or hospital when these ailments become severe or continuous. He also stated that they usually use orthodox drug because they believe it is the best as compared to traditional medicines. He also stated that there are known hereditary diseases such as hypertension in their family. He added that there is no history of mental health conditions and communicable disease in their family. Patient stated that his late father was diagnosed of hypertension before he died but both his grandfather and mother died of a natural death. There are no known drug or food allergy in their family as far as patient could recall, however, patient stated he is personally allergic to egg. He said, he throws up (vomits) each time he takes eggs. He affirmed that family members are currently in good health.

1.3 Socio-economic history

According to Hellmich, (2015), Socio- Economic history is the social science that studies how economic activity affects people and is shaped by social processes. In general, it analyzes how families progress, deteriorate, or regress because of their local economy.

During my interaction with the patient, he revealed that he has a very cordial relationship with the family and the people in his neighborhood. So, he disclosed that his family's source of finance is the money they generate from the farming. He earns an estimated GHS 1,000.00 every month from his farming work. Patient stated that, he and most of the rest of the family are beneficiaries of the National Health Insurance Scheme (NHIS), which serves as the main source of finance for their medical bills. Patient stated that there is no other financial support.

Due to the nature of their work, they are at risk of snake bites and injuries caused by farming tools.

The patient mentioned that there is no taboo in the family. Patients stated that most of the family members are Christians and they uphold values such as truth, respect and God-fearing life. They also attend social gatherings such as funerals, marriage and naming ceremonies in the community as it a core value in their culture. This implies that patient and the rest of the family have good social relationship with the community.

1.4 Patient developmental history

According to Weller, (2019) growth is the progressive development of a living thing, especially the process by which the body reaches its complete physical development.

Development refers to the process of growth and differentiation which involves cognitive, psychosexual and psychosocial processes (Weller, 2019).

Maturation is the process of becoming completely developed mentally or emotionally (Walter, 2013).

According to Mr. A. K. B. he was born per spontaneous vagina delivery in the house on 20th March, 1947. He was not able to state whether he was exclusively breast fed or not. He started sitting on the fourth month, crawling by the sixth month, standing and walking in the ninth month, as per the information given him by his late parents. He was not vaccinated against the childhood preventable disease as the vaccination had not begun when he was borne. There was no mark on his right deltoid muscle. He observed secondary sexual characteristics such as deep voice, broadening of shoulders, growing of pubic hair around the age of 15 years. According to Mr. A. K. B., he remembers ever starting school but cannot remember at what level he stopped. However, he could recall that he did not continue his education due lack of interest in school. Patient said he had the desire to be a professional footballer. Mr. A. K. B. is happily married with six children and believes he has been able to take good care of his children as either of them is able to live an independent live. He is very friendly and always support his friends around him in terms of advice, food and etc.

According to Erik Erikson (1902); Theory of Psychosocial Development, my client falls within “Integrity versus Despair” (65 years and above) which states; during this time that individuals contemplate their accomplishments and are able to develop *integrity* if they see themselves as leading a successful life. If they see their life as unproductive, or feel that they did not accomplish their goals, they become dissatisfied with life and develop *despair*, often leading to depression and hopelessness.

My interaction and observation on Mr. A. K. B. revealed he has developed a sense of integrity. Even though he desired to be a professional footballer which he could not accomplish, he has achieved so much in other aspects of life and had made each of his children independent. He therefore believes he would have peace after his death.

1.5 Patient lifestyle and hobbies

Life style is defined as the pattern of daily living that an individual develops (Weller, 2022).

Mr. A. k. B. usually wakes up around 6:00am in the morning and prays each morning before getting out of bed and goes to bed around 8:30pm in the evening.

He empties his bowel two time daily and empties his bladder for 4 times daily. He baths twice daily and maintains his oral hygiene twice daily. In the morning, he mostly eats porridge with bread as his breakfast. His favorite food is fufu with garden eggs soup. He takes three square meals daily. However, Mr. A. K. B. mentioned that he is not allergic to any drug or food with the exception of egg. He also states that he does not smoke but takes alcohol once in a while. Mr. A. K. B. stated that during week days, he goes to the farm, and on weekends too he rests and sometimes he attends funerals and weeding on Saturday. On Sunday, he prepares for church in morning and after church he goes to his friend's house for them to play some indoor games like "oware and dame". His hobbies are listening to news and watching football on the television. Mr. A. k. B. started that, he is the breadwinner of his family and he also started that he has interest in caring for both other family members and his grandchildren. He is very sociable and relates well with his family and neighbors. Patient stated that he sees to the welfare of the men's group in his church. He likes hardworking and generosity and dislikes immoral behaviors and laziness. My personal impression about him is that he very sociable, kind and generous.

1.6 Patient past medical history

Patient's past medical history involves patient's health status prior to the presenting problems he brought to the hospital.

According to Mr. A. K. B. he had no childhood illnesses such as; measles, whooping cough etc. as he grew from childhood through adolescence to adulthood. Mr. A. K. B. also said he has no known allergies to drugs, animals, insects or any food with the exception of egg. Mr.

A. K. B. has never been involved in any road traffic accident or major injuries before. According to Mr. A. K. B., he had been hospitalized a couple of times due to his hypertension. He sometimes suffers from mild headache and fever. He is normally put to complete bed rest and medications given until his health becomes quite normal and then discharge home. Patient has no physical disabilities.

1.7 Patient present medical history

Present medical history involves patient's actual problem that prompted him to come to the hospital. According to Mr. A. K. B., he was feeling well until 17th August, 2023. In the morning when he started experiencing headache, dizziness and tiredness when walking. He took tablet paracetamol 1g, but the signs and symptoms were still persisting. At 2:30pm Mr. A. K. B. was rushed to the emergency unit of the Holy Family Hospital, Berekum where his blood pressure was checked and recorded as 200/130mmHg, temperature 35.6°C, Pulse 83 beats per minute and Respiration 22 cycles per minute. He was monitored and managed at the emergency unit with IV furosemide 10mg stat, Nifedipine 20mg stat, then 30mg bd x 30, Methyldopa 500mg bd x 30. Mr. A. K. B.'s blood pressure was monitored every 4 hours for two days until he became stable, with blood pressure recording 147/87mmHg and the transferred to the male's ward on 19th August, 2023.

1.8 Admission of patient

Admission is the process of receiving a patient into the ward in order to ensure continuity of the nursing care to enhance the smooth and faster recovery and to prevent any complications from arising (McIntosh, 2013).

On 19th August, 2023, Mr. A. K. B. was accompanied by his daughter and two student nurses, myself inclusive to the male's ward from emergency unit per wheel chair, in a fully conscious state at 12:30pm. Upon arrival, patient and relative were welcomed to the ward and immediately

introduced to the nurses at the nurses' station. Patient was then made comfortable in an already made simple unoccupied bed and relative offered seat by the bedside. His hospital card was taken to confirm his admission to the ward. Patient had come to ward with complaints of headache, chest pain, dizziness and tiredness when walking and had been diagnosed of Hypertension.

On arrival, I checked and recorded his admission vital signs as follows;

Temperature	36.1°C
Pulse	71 beats per minute
Respiration	21cycles per minute
Blood Pressure	147/87mmHg

Mr. A. K. B.'s weight on admission was 65kg and was to be managed with;

1. Tablet Lisinopril 10mg daily x 30 days
2. Tablet Nifedipine 30mg bd x 30 days
3. Tablet Methyldopa 500mg bd x 30 days
4. Tablet Paracetamol 500mg bd x 5 days

Fasting Blood Sugar level (FBS) done earlier at the emergency unit recorded as 5.0 mmol/L.

Mr. A. K. B. and daughter were reassured of the competency of the medical team at the males' ward to do everything medically possible to help restore his health to normal. This was done to help relieve patient of any fear and to ensure trust in the medical team.

His particulars were entered into the admission and discharge book. Head to toe examination was performed to rule out any other possible health problem that is not stated by patient, and no other was observed. Patient and daughter were then introduced to all nurses on duty and the other patients at the ward. They were also oriented to the ward and its annexes. The hospital's policy

regarding the cost of care was explained to them, as National Health Insurance Scheme (NHIS) caters for some part of medical cost and the other part taken by the patient. Hospital policy regarding visiting time (which is 5:30am in the morning and 4:30pm in the evening) was also echoed to patient and daughter to be forwarded to the rest of the family that may want to visit. His daughter was informed about the importance of visiting her father regularly, because her involvement in the care will aid in early recovery of her father. Patient was reassured and made comfortable in bed. Plan of care was drawn to help curtail patient's problems on admission.

Continuous psychological care was given to patient so as to allay his fears. Therapeutic effects of medication as well as vital signs (especially blood pressure, respiratory rate and heart rate) were monitored regularly and noted in the nurses' notes. Patient was taught certain lifestyle modifications like avoidance of drinking alcohol, smoking and reduction of foods high in saturated fat and cholesterol as this may aggravate the condition.

I later introduced myself to them as a nursing student of Holy Family Nursing and Midwifery Training College, Berekum. A final year student with the desire to take Mr. A. K. B. as my client for patient/family care study so as to render individualized comprehensive nursing care until he is discharged. I made it known to them that it was a requirement by the Nursing and Midwifery Council of Ghana in partial fulfillment towards the award of license to practice as a Registered General Nurse. I further added that, they are allowed to withdraw their consent each time they feel so for whatever reason. They were very happy and agreed to my request and promised to co-operate fully in caring for Mr. A. K. B. They were enlightened about the details of the care study, not withholding the fact that three home visits would be made on three separate occasions to assess the home environment to ascertain how it might have contributed to his

condition. I assured them of privacy and confidentiality. Mr. A. K. B. was chosen because, I wanted to put into practice all the theoretical knowledge I have had regarding the management of hypertension. Discharge planning including how Mr. A. K. B. will take his drugs in the house, reduction of stress, dietary modification and the need to report early to health care facility in case of upsurge of the signs and symptoms were initiated.

1.9 Patient concept of illness

During my discussion with my client, he did not attribute his sickness to any forces but he said it is a natural phenomenon and believed that by God's grace and mercy, and if he effectively takes his medications, he would be well without any complication and his health will be improved. He was happy of the warm reception from the hospital staff and believed he will get better sooner and be discharged.

1.10 Literature review

Hypertension is defined by the Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC 7) as a systolic blood pressure greater than 140 mm Hg and a diastolic pressure greater than 90 mm Hg based on the average of two or more accurate blood pressure measurements taken during two or more contacts with a health care provider (Hinkle et al., 2022).

A blood pressure of less than or equal to 120/80mmHg, is defined as normal, 120-129/80-89mmHg as pre-hypertension and 140/90mmHg or higher as hypertension. The stage 2 hypertension is described as one that is above or equal to 160 for the systolic and 100 for the diastolic blood pressures (Tortora & Derrickson, 2022).

According to Tortora & Derrickson, (2022) one is considered to be hypertensive when one's average of three or more blood pressure readings taking at rest several days apart at about the same time of the day are persistently above 140/90mmhg. Hypertension is not diagnosed on a

single reading but based on several readings. It is the diastolic pressure that is very crucial in determining the degree of the hypertension.

Incidence

According to Tortora & Derrickson (2020) the elderly are usually prone to developing hypertension due to the aging process. Some children could also develop hypertension as a result of any congenital malformations. Also, hypertension is more common in males than in females. It is usually detected between the ages of 20 to 40 years.

Types of Hypertension.

According to Tortora & Derrickson (2020) hypertension can be grouped into primary (essential) hypertension and secondary hypertension.

Primary hypertension

Primary hypertension is also referred to as essential or idiopathic hypertension (Tortora & Derrickson, 2020). It is the commonest type of hypertension accounting for 90% of hypertensive cases. It has no identifiable cause and common among persons between the ages of 30 and 50 years. It is asymptomatic until complications occur. In Patients with primary hypertension, it is not possible to define a specific underlying cause.

According to Hinkle et al., (2022) risks factors associated with primary hypertension include;

1. Increasing age
2. Family history
3. Obesity or overweight
4. Alcoholism

5. Cigarette smoking
6. Anxiety
7. Emotional stress
8. Nutritional imbalance (too much saturated fat or sodium diet intake)

Secondary hypertension

According to Hinkle et al., (2022) secondary hypertension is distinguished from essential hypertension by having a known underlying cause. Common causes include the following;

Renal disease

A reduction of blood flow or a destruction of kidney structures causes hypertension.

Diseases such as nephritis, stenosis of the kidney and polycystic disease bring about changes in blood flow in the kidney structures to cause hypertension. With a reduced blood flow to the kidney, the kidney reacts by producing a proteolysis enzyme called renin. In the blood stream, rennin acts upon plasma protein to produce angiotensin I which is converted to angiotensin II. This has a vasoconstrictive effect that leads to increase in blood pressure. Renin also stimulates the adrenal glands to secrete aldosterone which causes water and sodium retention which increases the blood volume and consequently blood pressure.

2. Endocrine disorders

A tumour of the adrenal medulla (pheochromocytoma) secretes adrenaline that has a vasoconstrictive effect which results in a raise in the blood pressure. Increased aldosterone secretion also increases reabsorption of sodium and water to cause a raise in blood pressure. For example; in Cushing's disease, it increases secretion of adrenacorticoids causing hypertension.

3. Drugs

Drugs such as oral contraceptives are known to contain estrogen. In the liver, estrogen increase angiotensin to cause hypertension. If the drug is discontinued for 6 months, blood pressure returns to normal.

4. Pregnancy related diseases

In pregnancy certain diseases can bring about hypertension. Conditions like pre-eclampsia and eclampsia results in hypertension.

Other forms of hypertension

As cited by Hinkle et al., (2022) other forms of hypertension are;

1. **Malignant hypertension:** It is a severe form of hypertension. Malignant hypertension progresses rapidly and results in necrosis of the small arteries of the heart, kidneys, brain and eyes (target organ). Dysfunction of the organ ensues and without medical treatment the course of malignant hypertension is rapidly fatal. Most persons do not survive longer than two years. This condition is seen most often in black men under the age of 40. Patient may experience headache, seizures, papilloedema and retinal haemorrhage.
2. **Hypertensive emergency:** A situation in which blood pressure is severely elevated and there is evidence of actual or probable target organ damage.
3. **Hypertensive urgency:** A situation in which blood pressure is severely elevated but there is no evidence of target organ damage.
4. **Isolated systolic hypertension:** a condition mostly commonly seen in the elderly in whom the systolic pressure is greater than 140 mmHg and the diastolic pressure is within normal limits (less than 90 mmHg).

5. Rebound hypertension: blood pressure that is controlled with medication and that becomes uncontrolled (abnormally high) with the abrupt discontinuation of medication.

Causes of Hypertension

According to Hinkle et al., (2022) the causes of hypertension are as follows;

Decreased vasodilatation of the arterioles related to dysfunction of the vascular endothelium.

Increased renal absorption of sodium, chloride and water related to a genetic variation in the pathways by which the kidneys handle sodium.

Increased activity of the rennin- angiotensin- aldosterone system, resulting in expansion of the extracellular fluid volume and increased systemic vascular resistance.

Risk/Predisposing factors of Hypertension.

Hinkle et al., (2022) identified the following as risk factors of hypertension;

Modifiable factors

1. Alcoholism
2. Cigarette smoking
3. Anxiety
4. Emotional stress
5. Nutritional imbalance (too much saturated fat or sodium diet intake)
6. Drugs such as steroids and cocaine
7. Obesity

Non- modifiable factors

1. Increasing age
2. Family history

3. Gender
4. Hormonal factors

Pathophysiology

Blood pressure is the product of cardiac output multiplied by peripheral resistance. Cardiac output is the product of the heart rate multiplied by the stroke volume.

In normal circulation, pressure is transferred from the heart muscle to the blood each time the heart contracts, and then pressure is exerted by the blood as it flows through the blood vessels. Hypertension can result from an increase in cardiac output, an increase in peripheral resistance (constriction of the blood vessels), or both. Although no precise cause can be identified for most cases of hypertension, it is understood that hypertension is a multifactorial condition. Because hypertension is a sign, it is most likely to have many causes, just as fever has many causes. For hypertension to occur there must be a change in one or more factors affecting peripheral resistance or cardiac output. In addition, there must also be a problem with the body's control systems that monitor or regulate pressure.

Example, in renal ischaemia due to renal disease like nephritis, the kidney releases the enzyme renin. In the blood stream, renin acts upon a plasma protein angiotensinogen to produce angiotensin I. The angiotensin I is then converted to angiotensin II, by another enzyme called angiotensin converting enzyme produced in the pulmonary vasculature. The angiotensin II (a potent vasoconstrictor) causes widespread of peripheral resistance by

constricting blood vessels. This leads to increase in blood pressure. The angiotensin II also increases the secretion of aldosterone by the adrenal cortex leading to sodium and water retention which also increase the blood pressure. (Harding, Kwong, Roberts, Hagler, & Reinisch, 2020) revealed that; single gene mutations associated with the mechanisms used by the kidneys to reabsorb sodium ions have been identified for a few rare types of hypertension, but most types of hypertension are thought to be polygenic (i.e., mutations in more than one gene).

Clinical Manifestation

Hypertension is usually referred to as the “Silent killer” because it is frequently asymptomatic and usually detected on a routine physical examination of blood pressure (Hinkle et al., 2022).

Ministry of Health, Ghana (2017) outlined the following as clinical manifestations of hypertension;

1. Headaches
2. Palpitations
3. Dizziness
4. Easy fatiguability
5. Blood pressure $\geq 140/90$ mmHg
6. Epistaxis

Diagnosis

Ministry of Health, Ghana (2017) outlined the following as diagnostic investigations of hypertension;

- a. Full blood count to check haemoglobin level to rule out anaemia and WBC to rule out any infection.
- b. Urinalysis for protein to see if the kidneys are involved.
- c. Blood urea, electrolytes and creatinine level test is also done to evaluate the kidney's function.
- d. Fasting blood sugar to check patient's blood sugar level.
- e. Serum lipids
- f. Serum uric acid; Uric acid levels establish a baseline since the levels often rise with diuretic therapy.
- g. Chest X-ray to detect cardiomegaly.
- h. 12-lead ECG to find out the conducting activities of the heart.
- i. Ultrasound scan of kidneys and adrenals (in suspected secondary hypertension)
- j. Echocardiogram

Hinkle et al., (2022) added the following;

- a. Health history and
- b. Physical examination

Medical Management

The goal of hypertension treatment is to prevent complications and death by achieving and maintaining the arterial blood pressure at 140/90 mm Hg or lower (Hinkle et al., (2022)).

Pharmacological therapy

The medications used for treating hypertension decrease peripheral resistance, blood volume, or the strength and rate of myocardial contraction (Hinkle et al., (2022)).

Medications include;

Diuretic

Diuretic are not antihypertensive, but their combination with antihypertensive works effectively in lowering blood pressure. According to Hinkle et al., (2022) diuretic are grouped under the following heading with examples:

Thiazide Diuretics: These groups inhibit reabsorption of sodium chloride in distal convoluted tubule in the nephron Examples Bendrofluazide. This group (bendrofluazide) was prescribed to the patient.

Loop Diuretics: These groups of diuretic inhibit reabsorption of sodium and chloride in ascending loops of Henle. Examples include Furosemide (laxis), Torsemide etc.

Potassium Sparing Diuretic: these group of diuretics acts to block the effect of aldosterone on the renal tubule leading to loss of sodium and water and the retention of potassium.

Example includes, Spironolactone, Amiloride etc

Antihypertensive

Central Alpha 2 Agonist: These groups of drugs displace norepinephrine from their storage site. Examples include Methyldopa (Aldomet) and Clonidine

Beta – Blockers: These groups of drugs block the effect of catecholamine from receptor site to decrease cardiac output. Examples include Propranolol, Atenolol etc.

Alpha 1 Blocker: This drug work similar to vasodilators. They are peripheral vasodilators by acting directly on blood vessel. Examples include Terazosin, Prazosin etc.

Vasodilators: These drugs decrease peripheral resistance by dilating blood vessel. Examples include Hydralazine, Sodium Nitroprusside etc

Angiotensin Converting Enzyme Inhibitors: These groups of drugs lower blood pressure by blocking the conversion of angiotensin I to angiotensin II, a potent vasoconstrictor.

Examples include, Lisinopril and Captopril

Angiotensin II Receptor Blockers: These drugs also block the effect of angiotensin II at the receptor site. Examples include Losartan, Candesartan etc

Calcium Channel Blockers: These groups of drugs work effectively to lower blood pressure by inhibiting influx of calcium ion into the blood vessel. Examples; Nifedipine, Amlodipine etc

Direct Renin Inhibitors: they block the activities of enzyme renin. Example Aliskiren

Non - Pharmacologic Management

- I. Dietary changes
- II. Lifestyle modifications

Dietary Changes

- i. Sodium intake like salt should be restricted to reduce retention of water
- ii. Increase fiber intake to prevent constipation
- iii. Increase potassium intake such as banana as it increases extracellular potassium and sodium level
- iv. Caloric and fat restriction
- v. Adequate intake of fruits and vegetables

Lifestyle Modifications

- i. Regular exercise. Regular aerobic exercise such as jogging, walking and swimming can help control blood pressure. It can cause about 10mmHg decrease in systolic blood pressure.

- ii. Avoidance of cigarette smoking
- iii. Moderation of alcohol intake
- iv. Stress management

Nursing Management

As discussed by Hinkle et al., (2022), the patient is nursed under the following headings;

Psychological Management

The patient is reassured that with the good nursing interventions and medical team, the blood pressure will fall within normal range so far as he remains in the hospital. This is done to relax patient, win his cooperation and confidence and also to relieve and prepare her psychologically for what to expect.

The patient was encouraged to ask questions about his condition and any other misconceptions he has for clarification.

Patient was shown other recovering patient who suffered the same chronic disease condition

Patient was made free to operate within his capacity thus, engaging a little chat with bed mates to avoid excess thinking since it can mildly precipitate disease condition.

Positive aspect of the medical therapeutic treatment was centered on to patient.

Patient was taken through series of counselling sessions and multiple diversional therapies like, discussing the good olden days which he shared his experience, television watching and newspaper reading.

Patient and family should be made to understand that with their maximum co-operation, the condition can be managed.

Position

Put patient in an upright position to ensure breathing and to expand the chest supported with back rest and pillows.

The patient may also assume a comfortable and more suitable position.

Rest and sleep

Ensure enough rest and sleep to enhance relaxation, this is done to conserve energy, relax patient, promote healing, and reduce stress.

Patient should be given warm baths, proper ventilation and ensure that his bed is neatly laid and made free from creases.

Avoid unnecessary visitations during sleeping hours and carry all nursing activities at once to avoid sleep pattern interruptions.

Patient is given a complete bed rest for the first two weeks because of dizziness experienced by patient.

Observation

Vital signs such as Temperature, Pulse and Respiration should be monitored two hourly or four hourly to know the state or condition of patient, as to whether it is deteriorating or progressing and take appropriate interventions accordingly.

Patient is also observed for therapeutic effect and side effect of drugs as well as mental orientation of patient to time and place.

Monitoring of patient's intake and output chart is done and balanced at the end of 24hours.

Blood pressure monitoring is done every hourly until stable.

Observe for possible complication that can occur.

Personal hygiene

Patient may be assisted to take his or her bath twice daily or given a bed bath in order to remove dirt, microbes and sweat from the skin, improve circulation for comfort and relaxation.

Patient's is also assisted to shave beard. Hands and feet must also be cared for in order to prevent them from harbouring microbes or injuring patient.

Patient's gown must be changed regularly including bed linens and soiled diapers.

Oral care was done and sometimes assisted to do it.

Hand and feet are cared for by soaking them in water to soften it after which it trimmed to the likeness of patient.

Care of the mouth is done by use of toothbrush and paste, and in an unconscious state swap and normal saline is used.

Ensure hand washing with soap and water before and after eating and also after visiting the toilet to prevent infection.

Nutrition

Patient's nutritional level is met by serving patient with a well- balanced diet that is low in sodium; low carbohydrate to help reduce weight in obese patient, low protein, low fat to prevent hyperlipidemia, enough potassium supplement such as banana and enough roughage to help reduce constipation.

Patient should be encouraged to take in vitamin supplement or fruits and vegetables to boost the body's immune system.

Patient should therefore be involved in planning diet.

Meals served should be presented attractively and given in bits or at regular intervals.

Patient must have input and output.

Exercise

Patient is encouraged and assisted if there is the need to either undertake passive or active exercises to his or her tolerance levels, in order to improve circulation and to prevent complications such as joint stiffness, aid in peristalsis and to prevent constipation.

Assist patient to sit up in bed, walk around the bed and gradual turning in bed.

Elimination

Serve bed pan or assist patient to visit the toilet at patient's request and urinals served when necessary.

Encourage more fluids and roughages to soften stools.

When patient is unable to micturate, nursing measures such as opening tap water for it to run, applying warm compresses on the lower segment of the abdomen and catheterization carried out as ordered by the doctor in the extent that all other measures taken to get patient to micturates fails.

Patient and Family Teachings

The diagnosis of hypertension is usually unexpected and asymptomatic. Yet, once the diagnosis is made, the patient is asked to modify meal patterns and food choices, adopt daily exercise routine and adhere to the new medication with a variety of side effects.

They are advised to go for check-up for blood pressure to be checked.

Health Education

Health education is given on hypertension, taking into consideration the definition, the cause, treatment and most especially the dietary changes.

Patient should be educated on the drug regimen and importance of taking it appropriately to prevent any complication.

The chronicity of the disease condition and specific instruction concerning prescribed therapy should also be emphasized.

Patients who smoke must be educated on the effect of smoking on hypertension. A balanced between activity and relaxation should be touched on.

Because the condition is hereditary, other members of the family should be educated to go for regular checking of blood pressure.

Prevention of Hypertension

Hypertension is a life-threatening condition and as best to prevent its occurrence. A positive outlook towards health is reflected in the individual's lifestyle and habits. Health promotion focuses on educating the entire public to form a positive and more comprehensive attitude towards health.

Primary prevention

Health education is the most ideal action taken in primary prevention of hypertension.

These are the habit or lifestyle of the general public as the causes and effects of hypertension, likewise how the environment becomes a risk factor in the promotion of the disease are taught.

Primary prevention of hypertension includes;

Early identification of the condition and providing prompt and appropriate treatment. This is done through regular screening of individuals to detect any abnormality and if present, prompt doctor's attention for adequate treatment to be given.

Stress management by avoiding unhealthy arguments.

Weight reduction.

Moderation of alcohol intake.

Avoid smoking cigarette.

Regular physical exercise.

Fat and sodium.

Secondary Prevention

This has to do with, prevention of complications of the conditions by using drugs. It can also be achieved through lifestyle modification;

1. Report signs and symptoms early

Educate hypertensive patients to join societies and groups in the community to relieve stress or emotional disturbances

Educate patient to monitor blood pressure and record for early detection of hypertension record.

Advice patient on the need for adequate rest

Tertiary prevention

Tertiary production involves rehabilitation which focuses on assisting the patient to live independent life after the complication has occurred.

Surgical Management

Surgical management may become necessary in the case of tumors (pheochromocytoma) and sclerotic changes of the renal arteries which may be the cause of secondary hypertension. This surgical intervention is known as adrenalectomy.

Complications

As opined by Hinkle et al., (2022) if hypertension is not identified early for prompt and effective treatment, it results in complications. These complications usually relate to the

various organs and structures which are dependent to the heart. The organs commonly affected are;

1. Left ventricular hypertrophy; Sustained high BP increases the cardiac workload and produces left ventricular hypertrophy (LVH). Initially, LVH is a compensatory mechanism that strengthens cardiac contraction and increases CO.
2. Heart failure; HF occurs when the heart's compensatory mechanisms are overwhelmed and the heart can no longer pump enough blood to meet the body's demands. Contractility is depressed, and SV and CO are decreased.
3. Hypertensive Heart Disease; The mechanisms by which hypertension contributes to the development of atherosclerosis are not fully known. The "response-to-injury" theory of atherogenesis suggests that hypertension disrupts the coronary artery endothelium. This results in a rigid arterial wall with a narrowed lumen and may account for the high rate of CAD, angina, and MI.
4. Renal insufficiency and failure; renal disease results from ischemia caused by the narrowing of the renal blood vessels. This leads to atrophy of the tubules, destruction of the glomeruli, and eventual death of nephrons. Initially intact nephrons can compensate, but these changes may eventually lead to renal failure.
5. Hypertensive encephalopathy; Hypertensive encephalopathy may occur after a marked rise in BP if autoregulation does not decrease the cerebral blood flow. Autoregulation is a physiologic process that keeps cerebral blood flow constant despite fluctuations in BP. Typically, as pressure in the cerebral blood vessels rises, the vessels constrict to maintain constant flow. When BP exceeds the body's ability to autoregulate, the cerebral vessels suddenly dilate, capillary permeability increases, and cerebral oedema develops. This

produces a rise in intracranial pressure. If left untreated, patients can die quickly from brain damage. may occur after a marked rise in BP if autoregulation does not decrease the cerebral blood flow. Autoregulation is a physiologic process that keeps cerebral blood flow constant despite fluctuations in BP. Typically, as pressure in the cerebral blood vessels rises, the vessels constrict to maintain constant flow. When BP exceeds the body's ability to autoregulate, the cerebral vessels suddenly dilate, capillary permeability increases, and cerebral oedema develops. This produces a rise in intracranial pressure. If left untreated, patients can die quickly from brain damage.

1.11 Validation of data

Validation of data is defined according to Weller (2019), as the extent to which a data is measure, indicator or method of data collection possesses the quality of being sound or true, as far as it can be judge. Information about patient's condition was collected from the patient and direct observation. Other information was also collected from the patient's laboratory investigation results. This information, both the subjective and objective data when compared were not different from each other. The data collected were free from errors and biases and therefore valid.

CHAPTER TWO

ANALYSIS OF DATA

2.0 Introduction

Analysis of data is an aspect of the nursing process which deals with evaluation, interpretation and giving meanings to the data collected. It also refers to the act of determining the component part of a substance (Weller, 2019).

This chapter aims at analyzing the data collected on the patient condition by comparing data collected to standard. It helps to identify the patients for appropriate and lasting interventions to be rendered in order to improve the health status of the patient. Data analysis comprises of the following:

1. Comparison of data with standards
2. Patient/Family health problem
3. Patient/ family strength
4. Nursing diagnosis

2.1 Comparison of data with standards

Comparison is the fact of considering something similar or of equal quality to something else (McIntosh, 2013).

This has to do with data collected from Mr. A. K. B. with standard ones. It covers the following areas,

- A. Diagnostic Investigations
- B. Causes
- C. Signs and symptoms
- D. Treatment
- E. Complications

A. Diagnostic Investigations / tests

Diagnostic investigations are findings which are carried out to determine the health status of a person (McIntosh, 2013).

Several laboratory investigations were requested and done on Mr. A. K. B. to confirm his diagnosis and help in his treatment. These include;

1. Fasting blood sugar (19/08/23)
2. Hemoglobin level estimation (20/08/23)
3. Blood film for malaria parasites (20/08/23)
4. Lipid profile (20/08/23)

Table 1 below shows the comparison of diagnostic tests carried out on client and those listed in the Literature Review.

Table 1: Diagnostic investigations/test in literature review compared with those carried out on patient.

DIAGNOSTIC TEST OUTLINED IN LITERATURE REVIEW	DIAGNOSTIC TEST CARRIED OUT ON THE PATIENT
History and physical examination	His was taken and physical examination also conducted
Routine urinalysis.	Was not done for patient
Serum electrolytes and uric acid.	Was not done for patient
BUN and serum creatinine.	Was not done for patient
Fasting blood sugar levels.	Was not done for patient
Chest X-ray.	Was not done for patient

Serial blood pressure monitoring.	Blood pressure was serially monitored
Echocardiogram.	Was not done for patient
Serum lipid profile, cholesterol and triglycerides.	Lipid profile was done for patient
Electrocardiogram (ECG) - left ventricular hypertrophy.	Was not done for patient
Ultrasound scan of kidneys and adrenals	Was not done for patient
Full blood count	Was done

From the comparison of data above, it is obvious most of the investigations in literature were not done for my patient. This was because the very few performed for my patient (typical of continual blood pressure monitoring and physical examination) were enough to confirm the diagnosis.

Furthermore, some other investigations were performed for patient that are not outlined in the literature. These were done to rule out various possible concurrent conditions.

Table 2: Details of diagnostic investigations carried out on patient

Date	Specimen	Investigation	Results	Normal Values	Interpretation	Remarks
19/08/23	Blood	Hemoglobin level estimation (Full blood count)	11.8g/dl	Males (13-18g/dl) Females (12-16g/dl)	Patient hemoglobin level was not within normal range indicating that he was anemic	Though patient Hb was not normal, no treatment was ordered.
19/08/23	Blood	Lipid profile	Cholesterol 1.21mmol/L	< 1.03- > 1.55 mmol/L	Within normal range	No treatment was ordered
19/08/23	Blood	Blood film for malaria parasites	No malaria parasites seen	Negative	No malaria parasites seen indicating that patient has no malaria	No treatment was ordered
19/08/23	Blood	Fasting blood sugar	5.1mmol/dl	3.89-6.66 mmol/dl	Patient blood sugar level was within normal range indicating that he was not diabetic.	No treatment was ordered

B. Causes of Patient's Condition

Considering the factors that cause hypertension as indicated in the literature review, Mr. A. K. B. hypertension was as a result of advanced age. As we age, blood vessels become inelastic due to collagen fibers deposited in the walls and cannot expand to accommodate any increase in blood flow thereby causing an elevated blood pressure. Again, due to the nature of his work, stress could be a contributive factor since it triggers the sympathetic nervous system to activate vascular response which is typically associated with increased cardiac output and causes an elevation in blood pressure. Hereditary also plays a role in patient's condition because both of his parents also suffered from hypertension.

C. Clinical features/ signs and symptoms

Table 3 below shows comparison of clinical features.

Table 3: Comparison of clinical features in the literature review with those the patient exhibited

CLINICAL FEATURES IN LITERATURE REVIEW	CLINICAL FEATURES EXHIBITED BY MR. A. K. B.
Elevated blood pressure of 140/90mmHg or more.	Mr. A. K. B. had a blood pressure of 147/87mmHg on admission
Epistaxis	Mr. A. K. B. had no nasal bleeding
Dizziness	Mr. A. K. B. did complain of dizziness
Palpitations	Mr. A. K. B. had no palpitations
Fatigue	Mr. A. K. B. did not experience fatigue.
Headache	Mr. A. K. B. experience Headache

From the table above, patient experienced some clinical features as outlined in the literature review. Patient did not experience all the clinical features because he reported to the hospital quite early for treatment.

D. Specific medical treatment given to client

Treatment is the mode of dealing with a patient or disease, which aim at preserving and restoring injured parts by natural means (Weller, 2019).

1. Tablet Lisinopril 10mg daily x 30
2. Tablet Nifedepine 30mg bd x 30
3. Tablet Methyldopa 500mg bd x 30
4. Tablet Paracetamol 500mg bd x 5
5. IV Hydralazine 20mg stat

Table 4 below shows the treatment given to client compared with those in the Literature Review.

Table 4: Comparison of treatment outlined in the literature review with those given to my client.

TREATMENT OUTLINED IN LITERATURE REVIEW	TREATMENT GIVEN TO MY PATIENT
Diuretics	Was not administered
Adrenergic Inhibiting Agent	Tablet methyldopa was administered
Vasodilators	IV hydralazine was administered
Angiotensin converting enzyme inhibitor	Tablet lisinopril was administered
Calcium channel blockers	Tablet nifedipine was administered

Analgesics	Tablet paracetamol was administered
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From the table above, patient was given medications from most of the classes of drugs from the literature. This was aimed at bringing patient's blood pressure within normal as quickly as possible, so as to help restore his health.

Table 5: Pharmacology of drugs given to Mr. A. K. B.

Date	Drug	Dosage/route of administrations	Classification	Desired effect	Actual action observed	Side effects and interpretation
19/08/23	Tab Lisinopril	10mg dly×30days Oral	Angiotensin- Converting Enzyme (ACE) inhibitors	It relaxes the blood vessels.	High blood pressure reduced from 187/87mmHg to 130/90mmHg	Nausea, Blurred vision, Diarrhea, Headache. None was observed
19/08/23	Tab Methyldopa	500mg bd x 30 Oral	Central Alpha – 2 receptor agonist.	It relaxes and dilates (widens) blood vessels. Blood flows more freely and at a lower pressure.	Patient's blood pressure was reduced from 147/87mmHg to 130/90mmHg.	Bradycardia, dizziness. None was observed
19/08/23	Tab Nifedipine	30mg bd × 30days Oral	Calcium channel blocker	It lowers the blood pressure by relaxing the blood vessels	High blood pressure reduced from 147/87mmHg to 130/90mmHg	Dizziness, muscle cramp, headache and bloating or

				so the heart does not have to pump hard		swelling of the face, arms, and feet. None was observed
19/08/23	Tablet Paracetamol	500mg bd × 5days Oral	Antipyretic and Analgesic	Inhibition of prostaglandin synthesis or through an active metabolite influencing cannabinoid receptors.	Patient pain was relieved.	Liver damage, irritation to the GIT, swelling. None was observed.
20/08/23	Intravenous hydralazine	20mg stat intravenous	Vasodilators	It works by relaxing the blood vessels and increasing the supply of blood and oxygen to the heart while reducing its workload.	High blood pressure of 147/87mmHg reduced to 130/90mmHg	Headache, vomiting, constipation, diarrhea, loss of appetite, flushing and eye tearing. None was observed

E. Complications

With reference to the complication in the medical literature review, competent nursing care was provided and as a result Mr. A. K. B. did not develop any complication due to the prompt effective care rendered to him throughout hospitalization and home visit

2.2 Patient/family strength

According to Lewis, Dirksen, Heitkemper & Bucher, (2012), Strength is the quality of being strong. It also involves those that the family can also do to help in speedy recovery of the patient and those that the patient can perform.

The patient and family strength refers to attribute, characteristics or factors; tangible or intangible related to patient and family that facilitate nursing care of the patient. It included both related (specific) and unrelated (general) strength to patient's health problems. The following strength were identified:

1. Patient could tolerate pain medication. (19/08/2023)
2. Patient could cope with pain assessment. (19/08/2023)
3. Patient could verbalize the nature of the dizziness (19/08/2023)
4. Patient was in the right state of mind to help in educating him about his condition.
(19/08/2023)
5. Patient and family were ready to be educated on the new environment. (20/08/2023)

2.3 Patient health problems

Problem is defined as a situation, person that needs attention and needs to be dealt with or solved (McIntosh, 2013).

Health problems means medical ailment or an environmental condition that poses the risk disease or medical ailment. After the analysis collected from Mr. A. K. B., the following health problems were identified.

1. Patient complained of headache. (19/08/2023)
2. Patient complained of chest pain. (19/08/2023)
3. Patient complained of dizziness. (19/08/2023)
4. Patient appeared anxious. (19/08/2023)
5. Patient had inadequate knowledge about his condition. (20/08/2023)

2.4 Nursing diagnosis

According to NANDA International, nursing diagnosis is a clinical judgment concerning a human response to health conditions/life processes, or vulnerability for that response, by an individual, family, group, or community (Herdman & Kamitsuru, 2014).

Nursing diagnosis is based on patients complains and observation made by the nurse. It changes gradually as the patient responds to treatment from day to day. The formulated nursing diagnosis are as follows,

1. Headache related to elevated blood pressure (147/87). (19/08/2023)
2. Chest pain related to insufficient blood supply to the myocardium. (19/08/2023)
3. Dizziness related to inadequate cerebral tissue perfusion. (19/08/2023)
4. Anxiety related to unknown prognosis of disease condition. (19/08/2023)
5. Knowledge deficit related to lack of adequate information on the disease condition. (20/08/2023)

CHAPTER THREE

PLANNING OF PATIENT /FAMILY CARE

3.0 Introduction

According to Weller (2019), planning is the process in which the nurse and patient together consider the goals to achieve in meeting the patient's identified or potential problems in daily life and produce an individual care plan.

This chapter deals with the third phase of the nursing process which makes use of the nursing care plan to render service to the patient. This is where the nurse formulates nursing strategies and intervention to prevent, reduce or eliminate patient's health problems gathered during analysis.

In planning, objectives are set and prioritize in short- and long-term goals. Goals are set to develop upon and a plan of care are drawn to resolve the nursing diagnosis within a stipulated time frame.

The effectiveness of the care plan is highly dependent on good interpersonal relationship that exist between the medical staff and the patient as well as good nursing intervention.

3.1 Objectives/outcome criteria for the patient and family care

1. Patient's headache would be relieved within 24 hours as evidenced by;
 - a. Nurse observing a cheerful facial expression of patient
 - b. Patient verbalizing that his headache has been relieved
2. Patient chest pain would be relieved within 48 hours as evidenced by;
 - a. Nurse observing a cheerful facial expression of patient
 - b. Client verbalizing that he no more experiences chest pain
3. Patient would be relieved of dizziness within 48 hours as evidenced by;
 - a. Patient verbalizing that he does not feel dizzy any more

- b. Nurse observing patient move around the ward unassisted
- 4. Patient and relatives' anxiety would be relieved within 12 hours as evidenced by;
 - a. Nurse observing patient communicate freely with relatives without signs of apprehension.
 - b. Patient verbalizing relieve of anxiety.
- 5. Patient and family would gain adequate knowledge regarding cause, signs and symptoms, treatment and prevention of hypertension within 24 hours as evidenced by;
 - a. Nurse observing that patient and relatives are able to state at least two clinical manifestations or risk factors of hypertension.
 - b. Patient and relative being able to provide correct answer to questions posed to them with regards to the cause, treatment and prevention of hypertension

Table 6: Nursing care plan for Mr. A. K. B.

DATE AND TIME	NURSING DIAGNOSIS	OBJECTIVE/ OUTCOME	NURSING ORDERS	NURSING INTERVENTION	DATE AND TIME	EVALUATION	SIGN
19/08/2023 12:30pm	Headache related to elevated blood pressure (147/85).	Patient's headache would be relieved within 24 hours as evidenced by; a. Nurse observing a cheerful facial expression of patient	1. Reassure the patient 2. Assess patient's level of pain taking into consideration the onset, duration and intensity.	1. Patient was reassured of a competent nursing care to help relieve him of his headache 2. Assessment of patient's pain was done using pain numerical rating scale, taking into consideration the onset, duration and intensity and patient rated his pain 4.	20/08/2023 12:30pm	Goal fully met as patient stated relieve of his headache and also had cheerful facial expression.	S. D. G.

		b. Patient verbalizing that his headache has been relieved	3. Minimize surrounding noise. 4. Plan and carry out nursing care to ensure adequate bed rest. 5. Serve prescribed analgesic and anti-hypertensive drug	3. All categories of health care givers were advised to minimize noise and the volume of the radio and television sets were also reduced. 4. Nursing care was organised such that it minimised interruption of patient's rest to ensure adequate bed rest. 5. Prescribe analgesics (paracetamol) and anti-hypertensive drugs (nifedepine) were administered.			
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			6. Prepare a blood pressure chart for patient and monitor BP every 2 hours and record. 7. Document all procedures for continuity of care.	6. Blood pressure chart was prepared to monitor and record blood pressure every 2 hourly. 7. All procedures were documented in the nurses' notes for continuity of care.			
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Table 6: Nursing care plan for Mr. A. K. B.

DATE AND TIME	NURSING DIAGNOSIS	OBJECTIVE/ OUTCOME	NURSING ORDERS	NURSING INTERVENTION	DATE AND TIME	EVALUATION	SIGN
19/08/2023 12:40pm	Chest pain related to insufficient blood supply to the myocardium	Patient chest pain will be relieved within 48 hours as evidenced by; a. Nurse observing patient with a cheerful facial expression b. Patient verbalizing that he no more experiences chest pain.	1. Reassure the patient 2. Assess patient's level of pain taking into consideration the onset, duration and intensity. 3. Assess for factors that relieves the pain 4. Implement relieving factors	1. Patient was reassured of a competent nursing care to help relieve him of his chest pain. 2. Assessment of patient's pain was done using pain numerical rating scale, taking into consideration the onset, duration and intensity and client rated his pain 4. 3. Gentle massages on the chest were identified to relieve patient's chest pain. 4. Patient's chest was gently massaged regularly	21/08/2023 12:40pm	Goal fully met as patient verbalized that he nor more feels pain in the chest and also displayed a cheerful facial expression.	S. D. G.

			of patient's pain 5. Plan and carry out nursing care to ensure adequate bed rest. 6. Serve prescribed anti- hypertensive drugs 7. Document all procedures for continuity of care.	to help relive him of the pain 5. Nursing care was organized such that it minimized interruption of patient's rest to ensure adequate bed rest. 6. Prescribe anti- hypertensive drugs such as (lisinopril) was administered. 7. All procedures were documented in the nurses' notes for continuity of care.			
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Table 6: Nursing care plan for Mr. A. K. B.

DATE AND TIME	NURSING DIAGNOSIS	OBJECTIVE/ OUTCOME	NURSING ORDERS	NURSING INTERVENTION	DATE AND TIME	EVALUATION	SIGN
19/08/2023 12:50pm	Dizziness related to inadequate cerebral tissue perfusion.	Patient would be relieved of dizziness within 48 hours as evidenced by; a. Patient verbalizing that he does not feel dizzy any more b. Nurse observing patient move around the ward unassisted	1. Reassure patient. 2. Encourage patient to use walking aid like Zimmer frame or walker when moving from one area to another. 3. Encourage patient to have enough rest to help stabilize his blood pressure 4. Monitor vital signs especially blood pressure regularly.	1. Patient was reassured that measures will be taken to assist him regain his mobility. 2. Patient was encouraged and assisted to use Zimmer frame when moving from one area to another. 3. Comfortable bed was made for patient to promote enough rest to help stabilize his blood pressure 4. Blood pressure chart was made to monitor	21/08/2023 12:50pm	Goal fully met as patient verbalized that he no more feels dizzy and nurse observed patient moved about the ward unassisted.	S. D. G.

			5. Serve prescribed antihypertensive drug. 6. Document all procedures for continuity of care.	patient's blood pressure every 2 hourly. 5. Prescribed nifedepine was administered 6. All procedures were documented in the nurses' notes for continuity of care.			
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Table 6: Nursing care plan for Mr. A. K. B.

DATE AND TIME	NURSING DIAGNOSIS	OBJECTIVE/ OUTCOME	NURSING ORDERS	NURSING INTERVENTION	DATE AND TIME	EVALUATION	SIGN
19/08/2023 1:00pm	Anxiety related to unknown prognosis of the disease condition	Patient and relatives’ anxiety would be relieved within 12 hours as evidenced by; a. Nurse observing patient communicate freely with relatives without signs of apprehension b. Patient verbalizing	1. Reassure patient. 2. Encourage patient to voice out fears 3. Teach patient of relaxation techniques. 4. Educate patient on the condition	1. Patient was reassured that, he is in the care of competent healthcare team that would help to restore his health status. 2. Patient was encouraged to voice out fears and he stated he is uncertain if the condition was going to end his life. 3. Patient was taught of relaxation techniques like deep breathing exercises and encouraged to practice 4. Condition was explained into details	20/08/2023 1:00am	Goal fully met as patient verbalized that he no more feels anxious and nurse observed patient communicated freely with relatives and fellow patients without any apprehension.	S. D. G.

		relieve of anxiety	5. Encourage patient to listen to favorite programs on the radio to divert his mind from the anxiety. 6. Explain the need for hospitalization to patient and his family. 6. Document all procedures for continuity of care.	including available management to help clear his fears. 5. Patient was encouraged to listen to favorite programs on the radio to divert his mind from the anxiety. 6. The need for hospitalization was explained to patient and his family. 6. All procedures were documented in the nurses' notes for continuity of care.			
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Table 6: Nursing care plan for Mr. A. K. B.

DATE AND TIME	NURSING DIAGNOSIS	OBJECTIVE/ OUTCOME	NURSING ORDERS	NURSING INTERVENTION	DATE AND TIME	EVALUATION	SIGN
20/08/2023 8:00am	Knowledge deficit related to lack of adequate information on the disease condition.	Patient and family would gain adequate knowledge regarding cause, signs and symptoms, treatment and prevention of hypertension within 24 hours as evidenced by; a. Nurse observing that patient and relatives are	1. Reassure patient/family. 2. Assess patient and his family's level of knowledge to clarify any possible misconception about hypertension. 3. Create a conducive environment for learning by minimizing noise and	1. Patient and family were reassured that, adequate information would be provided to them for them to gain adequate knowledge on the condition. 2. Patient and his family's level of knowledge was assessed to clarify any possible misconception about hypertension. 3. A conducive environment was created for learning by minimizing noise and	21/08/2023 8:00am	Goal fully met as patient was able to answer simple questions posed to him regarding the condition.	S. D. G.

		able to state at least two clinical manifestations or two risk factor of hypertension. b. Patient and relatives being able to provide correct answers to questions posed to them with regard to the causes, treatment and prevention of hypertension	putting away all forms of destructors (example putting client's mobile phone on silence). 4. Create inter-personal relationship with client. 5. Ask questions about their health status. 6. Educated patient and family on the causes, clinical manifestations, as well as the treatment of hypertension.	putting away all forms of destructors; thus patient's was encouraged to put his mobile phone on silence during the period of education. 4. Inter-personal relationship was created by way of sitting close to patient's bed side, this was aimed at facilitating active listening. 5. Patient and family were asked about their general health status. 6. Patient and family were educated on the causes, clinical manifestations, as well as the treatment of hypertension.			
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			7. Document all procedures for continuity of care.	7. All procedures were documented in the nurses' notes for continuity of care.			
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CHAPTER FOUR

IMPLEMENTING PATIENT/FAMILY CARE

4.0 Introduction

Implementation is the actualization of the nursing care plan through nursing intervention Hinkle et al. (2022). Implementation is the fourth stage of the nursing process. It deals with the performance or putting in action the nursing interventions described in the nursing care plan. This stage deals with the actual care that was rendered to the patient and family from admission to the time of discharge.

4.1 Summary of actual nursing care

Nursing care rendered to Mr. A. K. B. started from the day of his admission (19/08/23) and continued till the termination of care. The summary of the care given was written daily as below.

DAY OF ADMISSION (19TH AUGUST, 2023)

On 19th August, 2023, Mr. A. K. B. was accompanied by his daughter and two student nurses, myself inclusive to the male's ward from emergency per wheel chair, in a fully conscious state at 12:30pm. Upon arrival, patient and relative were welcomed to the ward and immediately introduced to the nurses at the nurses' station. Patient was then made comfortable in an already made simple unoccupied bed and relative offered seat by the bedside. His hospital card was taken to confirm his admission to the ward. Patient had come to ward with complaints of headache, dizziness and tiredness when walking and had been diagnosed of Hypertension. He also looked anxious on observation.

On arrival, I checked and recorded his admission vital signs as follows;

Temperature	36.1°C
Pulse	71 beats per minute

Respiration 21cycles per minute

Blood Pressure 147/87mmHg

Mr. A. K. B.'s weight on admission was 65kg and was to be managed with;

1. Tablet Lisinopril 10mg daily x 30
2. Tablet Nifedepine 30mg bd x 30.
3. Tablet Methyldopa 500mg bd x 30.
4. Tablet Paracetamol 500mg bd x 5

Fasting Blood Sugar level (FBS) done earlier at the emergency unit recorded as 5.0mmol/L.

Mr. A. K. B. and daughter were reassured of the competency of the medical team at the males' ward to do everything medically possible to help restore his health to normal. This was done to help relieve patient of any fear and to ensure trust in the medical team.

His particulars were entered into the admission and discharge book. Head to toe examination was performed to rule out any other possible health problem that is not stated by patient, and no other was observed. Patient and daughter were then introduced to all nurses on duty and the other patients at the ward. They were also oriented to the ward and its annexes. The hospital's policy regarding the cost of care was explained to them, as National Health Insurance Scheme (NHIS) caters for some part of medical cost and the other part taken by the patient. Hospital policy regarding visiting time (which is 5:30 am in the morning and 4:30 pm in the evening) was also echoed to patient and daughter to be forwarded to the rest of the family that may want to visit. His daughter was informed about the importance of visiting her father regularly, because her involvement in the care will aid in early recovery of her father. Patient was reassured and made comfortable in bed. Plan of care was drawn to help curtail patient's problems on admission.

Continuous psychological care was given to patient so as to allay his fears. Therapeutic effects of medication as well as vital signs (especially blood pressure, respiratory rate and heart rate) were monitored regularly and noted in the nurses' notes. Patient was taught certain lifestyle modifications like avoidance of drinking alcohol, smoking and reduction of foods high in saturated fat and cholesterol as this may aggravate the condition.

I later introduced myself to them as a nursing student of Holy Family Nursing and Midwifery Training College, Berekum. A final year student with the desire to take Mr. A. K. B. as my client for patient/family care study so as to render individualized comprehensive nursing care until he is discharged. I made it known to them that it was a requirement by the Nursing and Midwifery Council of Ghana in partial fulfillment towards the award of license to practice as a Registered General Nurse. I further added that they are allowed to withdraw their consent each time they feel so for whatever reason. They were very happy and agreed to my request and promised to co-operate fully in caring for Mr. A. K. B. They were enlightened about the details of the care study, not withholding the fact that three home visits would be made on three separate occasions to assess the home environment to ascertain how it might have contributed to his condition. I assured them of privacy and confidentiality. Mr. A. K. B. was chosen because, I wanted to put into practice all the theoretical knowledge I have had regarding the management of hypertension.

Discharge planning including how Mr. A. K. B. will take his drugs in the house, reduction of stress, dietary modification and the need to report early to health care facility in case of upsurge of the signs and symptoms were initiated.

On admission at 12:30pm, patient complained of headache. A nursing diagnosis of "Headache related to elevated blood pressure" was formulated and an objective was set to

relieve patient of headache within 24hour. The following nursing interventions were implemented to help curtail patient's problem; patient was reassured of a competent nursing care to help relieve him of his headache, assessment of patient's pain was done using pain numerical rating scale, taking into consideration the onset, duration and intensity and patient rated his pain 4, all categories of health care givers were advised to minimize noise and the volume of the radio and television sets were also reduced, nursing care was organized such that it minimized interruption of patient's rest to ensure adequate bed rest, prescribe analgesics (paracetamol) and anti-hypertensive drugs (nifedipine) were administered, blood pressure chart was prepared to monitor and record blood pressure every 2 hourly and all procedures were documented in the nurses' notes for continuity of care.

Patient did have chest pain too on admission. So at 12:40pm, a nursing diagnosis of "Chest pain related to insufficient blood supply to the myocardium" was made. Ana objective was set to help relieve patient of chest pain within 48 hours. Various nursing interventions were done such as; patient was reassured of a competent nursing care to help relieve him of his chest pain, assessment of patient's pain was done using pain numerical rating scale, taking into consideration the onset, duration and intensity and client rated his pain 4, patient's chest was gently massaged regularly to help relive him of the pain, nursing care was organized such that it minimized interruption of patient's rest to ensure adequate bed rest, prescribe anti-hypertensive drugs such as (lisinopril) was administered and all procedures were documented in the nurses' notes for continuity of care.

By 12:50pm, patient was seen haven difficulty moving about at the ward. Upon further enquiries, patient stated he felt dizzy. A nursing diagnosis of "Dizziness related to inadequate cerebral tissue perfusion" was made. An objective was set to help make patient

free from dizziness within 48 hours. Interventions such as; patient was reassured that measures will be taken to assist him regain his mobility, patient was encouraged and assisted to use Zimmer frame when moving from one area to another, comfortable bed was made for patient to promote enough rest to help stabilize his blood pressure, blood pressure chart was made to monitor patient's blood pressure every 2 hourly, prescribed nifedipine was administered and all procedures were documented in the nurses' notes for continuity of care were implemented to help relieve patient of headache.

During the initial interaction with patient, it was observed that he showed signs of anxious. Further enquiries were made and patient confirmed that he has fear as he does not know what the outcome of the situation would be.

Therefore, a nursing diagnosis of "Anxiety related to unknown prognosis of the disease condition" was formulated. An additional objective was set to help relieve patient of anxiety within 12 hours at 1:00pm. Interventions were implemented to help meet set objective. Such interventions included; patient was reassured that, he is in the care of competent healthcare team that would help to restore his health status, patient was encouraged to voice out fears and he stated he is uncertain if the condition was going to end his life, patient was taught of relaxation techniques like deep breathing exercises and encouraged to practice, condition was explained into details including available management to help clear his fears, patient was encouraged to listen to favorite programs on the radio to divert his mind from the anxiety, the need for hospitalization was explained to patient and his family, and all procedures were documented in the nurses' notes for continuity of care.

Patient vital signs were checked and recorded at 2pm as in the appendix. Patient was served with "ampesi" and "kontomire" stew for lunch. He was made comfortable in bed to relax.

Patient was handed over to the night nurses around 8:20 pm and I said goodbye to him before leaving the ward.

Patient had his supper at 5:40pm. Due medications were administered at 6pm, personal hygiene was ensured and patient was handed over to the night nurses for continuity of care.

SECOND DAY OF ADMISSION 20TH AUGUST, 2023.

On 20th of August 2023, according to the night nurses, patient was awake during some hours of the night. The objective set on 19/08/2023 to help patient be relieved of anxiety was evaluated at 1:00am, and goal was fully met as patient verbalized that he no more feels anxious and nurse observed patient communicated freely with relatives and fellow patients without any apprehension. He woke up as early as 5:40am. His bed linen and clothes were changed to make him comfortable. He brushed his teeth and emptied his bowel, bladder and proceeded to take his bath.

At 6am, routine vital signs were checked and recorded as follows:

Temperature	35.4 degree Celsius (°C)
Pulse	68 beats per minutes (bpm)
Respiration	20 cycle per minute (cpm)
Blood Pressure	150/90 millimeters of Mercury mmHg

All due medication were also administered at 6:00am and recorded in the nurse's continuation Sheet as well. He took porridge and bread as breakfast.

During my interaction with patient, I realized that patient had very little knowledge on the condition.

In view of that, a nursing diagnosis of "Knowledge deficit related to lack of adequate information on the disease condition" was formulated at 8:00am. Further objective was set

to help patient and family gain adequate knowledge regarding cause, signs and symptoms, treatment and prevention of the condition within 24 hours. Various interventions carried out included; patient and family were reassured that, adequate information would be provided to them for them to gain adequate knowledge on the condition, patient and his family's level of knowledge was assessed to clarify any possible misconception about hypertension, a conducive environment was created for learning by minimizing noise and putting away all forms of destructors; thus patient's was encouraged to put his mobile phone on silence during the period of education, inter-personal relationship was created by way of sitting close to patient's bed side, this was aimed at facilitating active listening, patient and family were asked about their general health status, patient and family were educated on the causes, clinical manifestations, as well as the treatment of hypertension, and all procedures were documented in the nurses' notes for continuity of care.

At 8:30am patient was reviewed by Dr. A., and he ordered treatment to be continued, he however prescribed IV hydralazine 20mg stat. This was quickly obtained from the pharmacy and administered. The following laboratory investigations were also requested;

1. Fasting blood sugar
2. Hemoglobin level estimation
3. Blood film for malaria parasites
4. Lipid profile

Patient had a cheerful time with fellow patients, and family during his visiting hours and rested afterward. His vital signs were checked and recorded again at 10:00am and he laid in supine position to have some rest.

At 12:10 pm, Mr. A. K. B. lunch was served, he was served with Gari and Beans.

At 12:30pm, I evaluated the goal set on 19/08/2023 to help relieve patient of headache, and goal was fully met as patient stated relieve of his headache and also had cheerful facial expression.

At 2:00pm, his routine vital signs were checked and recorded. After the vitals, I observed patient had cheerful facial expression. I engaged him in conversation and took that opportunity to inform him of my intention to visit their house the following day. He was served with yam and stew for supper. He ate about half of it after which he was encouraged to watch television.

Mr. A. K. B. took rice and stew as super and took some oranges after that around 5:00pm.

At 6:00pm, his vital signs were checked and recorded, and all due medications were served and recorded in the treatment sheet and nurses' notes. All procedures and findings were recorded in the nurses' continuation sheet.

Patient took his bath around 7:00pm using warm water and he was handed over to the night nurses around 8:00pm.

THIRD DAY ON ADMISSION 21ST AUGUST, 2023

Mr. A. K. B. woke up at around 5:00 am according to the night nurses. He looked cheerful and active than he did on admission. He had slept for about 7 hours over the night. He maintained his personal hygiene that is brushing his teeth and bathing. His bed linen and clothes were changed to make him comfortable.

His vital signs at 6:00am were:

Temperature	36.0 °C
Pulse	76 bpm
Respiration	22 cpm

Blood pressure 150/98 mmHg

All due medication were also administered at 6:00am and recorded in the nurses' notes and treatment sheets. He was served with warm Milo beverage and bread for breakfast. Through conversation with patient at 8:00am, he revealed that he had a very sound sleep as reported by the night nurses earlier. Also, patient was able to answer some simple questions posed to him concerning his condition, and he was able to answer. Therefore, the objective set on the 20/08/2023 to patient gain adequate knowledge on the condition was evaluated. The goal was fully met as patient was able to answer simple questions posed to him regarding the condition.

Patient was reviewed by Dr. A. and asked to continue treatment and monitor client's blood pressure 4 hourly.

At 10:00 am, his vital signs were checked and recorded as his blood pressure was recorded as 160/90mmHg.

By 12:40pm, the goal set on 19/08/2023 to help relieve patient of chest pain was evaluated and was fully met as patient verbalized that he no more feels pain in the chest and also displayed a cheerful facial expression.

Mrs. A. K. B. was served with rice and vegetable stew for lunch. He ate almost all the food served.

At 12:50pm, objective set on 19/08/2023 to help relieve patient of dizziness was evaluated. The goal was fully met as patient verbalized that he no more feels dizzy and nurse observed patient moved about the ward unassisted.

After his lunch he was made comfortable in bed. His vital signs were checked and recorded around 2:00pm and his due medications were served and recorded in the nurses' note.

I informed patient of my plan to visit their home on this day which had already been discussed on the previous day. At 4:30 pm, I went to their house following the directions given to me by patient.

He was visited by some relatives. He looked cheerful and happy to see them. After that he ate fufu with light soup and dried fish as supper.

At 6:00pm, his vital signs were checked and recorded, and his due medications were served and recorded.

Patient took warm bath which was his preference at 6:55pm. All procedures and findings were recorded in the nurses' continuation sheet, and he was finally handed over to the night nurses around 8:00pm.

FOURTH DAY OF ADMISSION 22ND AUGUST, 2023

On 22nd August, 2023, Mr. A. K. B. woke up around 5:30am, he brushed his teeth and emptied his bowel before he took his bath. It was reported by the night nurses that he had a proper sleep.

At 6am, routine vital signs were checked and recorded as follows:

Temperature	36.5°C
Pulse	74 bpm
Respiration	24 cpm
Blood Pressure	140/90 mmHg

All due medication were also served at 6:00am and recorded in the nurse' continuation sheet as well. He was served with "Hausa koko" and bread as breakfast.

At 8:30 am, patient was reviewed by Dr. A., and he ordered treatment to be continued, with a possible discharge the next day.

At 10:00am, client's blood pressure was checked and recorded in the nurses' note as 150/100mmHg and all his due medications were served.

He was served fried yam with fish as lunch at 12:12pm and at 2:00pm, routine vital signs were checked and recorded again. He was served with yam and stew for supper.

At 6:00pm, his vital signs were checked and recorded, and all due medication was served and recorded in the treatment sheet and nurses' notes.

Patient took a warm bath. All procedures and findings were recorded in the nurses' continuation sheet. He was finally handed over to the night nurses.

FIFTH DAY OF ADMISSION/DAY OF DISCHARGE 23RD AUGUST, 2023.

On 23rd August 2023, Mr. A. K. B. woke up around 5:50am, he brushed his teeth.

He took his bath. He was observed having uninterrupted sleep.

At 6am, routine vital signs were checked and recorded as follows:

Temperature	36.2 °C
Pulse	70 bpm
Respiration	20 cpm
Blood Pressure	135/89 mmHg

All due medication were also served at 6:00am and recorded in the nurse' continuation sheet as well. He was served "lipton" and bread as breakfast.

At 8:30am patient was reviewed by Dr. A., and he ordered patient to be discharge as he was well and presented with no complains. Patient and relatives were informed about the discharge. His vital signs were checked and recorded as temperature- 36.4 degree Celsius, respiration- 19 cycle per minute, pulse- 66 beat per minute and blood pressure- 130/90 millimeter of mercury. Patient's daughter was made aware and the necessary preparations

for discharge started. His health care card was sent to the Billing Office for assessment and payment of his bills. No amount of money was paid for since those bills which were not covered by National Health Insurance Scheme had already been paid for in the previous days. Patient and relatives were assisted to pack their belongings. Patient was educated on nutritious diet low in fat, as well as maintaining good personal hygiene. Mr. A B. K. was discharged on his antihypertensive medications. Patient and relatives were educated to avoid self-medication and the need to come to hospital for proper treatment when sick. Education was given on the dosage and the storage of the medication. Also, education was given on the adverse effects of the medications and that they should report if any adverse reactions are observed. Patient and relatives were also educated on the importance of taking the drugs. They were then informed about the importance of review and follow up care and to report promptly to the hospital if any member of the family falls sick for treatment to prevent complications. Patient and relatives were also educated on his dietary restrictions especially on the fact that he should reduce the intake of salt and fatty diet and if possible, avoid them totally. He was also advised to eat balanced diet, not to eat very late in the night and to avoid sleeping right after eating. Patient and relatives were educated to eat a lot of fruits and vegetables. Patient and relatives were also educated on personal hygiene of which he was encouraged to make hand washing with soap and water a habit.

Terminal disinfection of the bed and the linen was done to prevent cross-infection to other patient and the bed was made ready to admit another patient. The date for review was communicated to them, which was on 8th September, 2023. Her particulars were entered in the admission and discharge book and daily ward state. I told patient and about my next home visit on 30th August, 2023. Patient and the family bid the rest of the patients and staff

goodbye. I accompanied them to the hospital gate where they boarded a tricycle and bid each other a good bye.

4.2 Preparation of patient/family for discharge and rehabilitation

Preparation of patient/family for discharge started on the day of admission, 19th August, 2023 and continued till the day of discharge. They were told that the hospital is a temporal place for them and that they will be discharged if patient's health is restored. The reason was to enable patient's family to take active role in his care for speedy recovery and also to give an insight into his condition. On the day of discharge (23rd August, 2023), emphasis was made on the need to visit the hospital immediately anytime he falls sick to promote early detection and treatment to avoid complications. Patient and relatives were educated to avoid self-medication and the need to come to hospital for proper treatment when sick. Education was given on the dosage and the storage of the medication. Also, education was given on the adverse effects of the medications and that they should report if any adverse reactions are observed. Patient and relatives were also educated on the importance of taking the drugs. They were then informed about the importance of review and follow up care and to report promptly to the hospital if any member of the family falls sick for treatment to prevent complications. Patient and relatives were also educated on his dietary restrictions especially on the fact that he should reduce the intake of salt and fatty diet and if possible, avoid them totally. He was also advised to eat balanced diet, not to eat very late in the night and to avoid sleeping right after eating. Patient and relatives were educated to eat a lot of fruits and vegetables. Patient and relatives were also educated on personal hygiene of which he was encouraged to make hand washing with soap and water a habit.

4.3 Follow- up/home visits and continuity of patient/family care.

A home visit is a touring to the home of the patient with the aim of promoting health through education and assessment of health status. It is carried out before and after discharge.

The reason for this home visit is to help assess the nature of the patient and family's home/community and the people in the home/community to determine people at risk (vulnerable) of contracting the disease. It also helps patient's family to be educated on any unhealthy living and factor that will be identified. State of patient and family's health are assessed and documented.

FIRST HOME VISIT (21/08/23)

On 21st August, 2023, I made my first home visit to my patient's house, the aim was to verify the data I had already collected, to check if patient's environment contributed to his illness and to give health education on any health issue identified in the house. I took off from the ward, around 4:30pm and reached there at exactly 4:44pm.

The visit was arranged with the patient's wife. Their house is closer to Senase R/C primary and J. H. S., which made it easy to locate. On arrival, I was warmly welcomed, and introduction was made after exchanging greetings with patient's children. I was offered a seat and a cup of water. I introduced myself as a student nurse and a member of the health team who were caring for Mr. A. K. B. at Holy Family Hospital, Berekum and they were glad to see me. The reason for the visit was made known to them. With permission, the environment of the house was observed, and a quick observation was made about the environment. There are 3 rooms, a kitchen, bathroom, and a water closet toilet. It was a clean environment. I then took my time to educate them on the need to continue practicing

personal and environmental hygiene to prevent mosquitoes from breeding, sleep under treated net and also to wash hand with soap and water before and after eating.

They were very much grateful and thanked me. I also thanked them and asked for permission to leave which they agreed and informed them that there would be another visit to Mr. A. K. B. after he is discharged. I left the house at 5:37pm.

SECOND HOME VISIT (30TH AUGUST, 2023)

My second home visit was carried out on 30th August, 2023. This was a week after patient's was discharge. The purpose of the visit was to assess his response to treatment at home and find out if patient and family were adhering to the dietary and other health advice given to him while he was on admission and also to remind him of the date of review. I got to the house around 3:30pm. I was received warmly once again by patient and his relatives. They were happy to see me again. I greeted them and asked of their health. They all said they were fine. I observed that the surroundings were clean, the place looked dry, and their refuse was burnt. I expressed my happiness about their appreciation of my recommendations to them and they were grateful.

I reminded patient and relatives about his date of review which was 8th September, 2023. I congratulated them for their efforts towards maintaining good health.

I then took the opportunity to talk to them about the condition again, stressing on the causes, predisposing factors and the dietary and lifestyle restrictions. I told him to avoid stressful situations, eating fatty diet, taking alcoholic beverages and too much salt, and also avoid eating too late in the night. I also encouraged him to continue to take his drugs, come for the review and visit the hospital anytime he does not feel well. I encouraged him to have enough rest and eat a lot of fruits and vegetables.

I informed patient and relatives that I will hand over the care to a community health nurse next time I visit. I told him that it is necessary to ensure continuity of care. I finally asked for permission to leave with a promise that I would visit them lastly to end our professional encounter, which was granted. I then left patient house at 5:00pm back to my house.

REVIEW DATE: 1ST SEPTEMBER, 2023

On the day of review, the patient was met at the out-patient's department (OPD) at 8:00am. We exchanged greetings and he was welcomed. I collected his folder from records and sent it to the consulting room. His vital signs were checked and recorded as follows:

Temperature	36.1 °C
Pulse	76 bpm
Respiration	19 cpm
Blood Pressure	127/85 mmHg

The doctor examined him. During review, no complain was given by patient. He was advised to adhere to the medications as prescribed and also much emphasis was placed on the reduction of salt intake. Patient was accompanied to the hospital gate where he boarded a taxi, and I bade him goodbye.

THIRD HOME VISIT-11TH SEPTEMBER, 2023

The last home visit was made on the 11th September, 2023. I took off from my house at 4:00pm and reached there at exactly 4:15pm. The main aim of the visit was to find out how Mr. A. K. B. and his family members were doing after review and to terminate the care by introducing them to the community health nurse who would continue with the care.

Patient and family were happy on seeing me in the company of the Community Health Nurse Ms. H. They told me they were all fine with no complains. They were educated on

the need for periodic medical check-up, stress reduction, drug adherence and dietary regimen.

I therefore introduced the Community Health Nurse to patient and family who promised to do the follow up visit and give any health information which would be needed. The family and patient also assured me of their cooperation with the community health nurse. I thanked them for the opportunity offered me to write a report on the care rendered to them.

Though it was a difficult task terminating the care with the patient and family, I did my best to make them aware that I was just ending the professional interaction, but I am always at their service whenever the need arises and would pass by to check on them occasionally.

They expressed their gratitude about the knowledge they have acquired on hypertension and the measures they have to put in to prevent the disease from occurring. I thanked them for their cooperation and sought permission to leave which was granted by patient and relatives.

I left patient's house around 5:20pm.

CHAPTER FIVE

EVALUATION OF CARE RENDERED TO PATIENT /FAMILY

5.0 Introduction

According to Hinkle et al., (2022) evaluation is the outcome of nursing actions against the anticipated goals and it is the final step in the nursing process. Evaluation in nursing care seeks to measure the effectiveness of assessment, diagnoses and implementation. Patient's health status is compared to goals of health care to determine goals achieved. It involves the members of the health team, patient and her family. Unachieved goals of nursing care plan are amended and care is terminated afterwards with conclusions made on the care rendered.

5.1 Statement of evaluation

During hospitalization, many goals were set with specific outcome criteria for the patient. They were aim at providing patient with holistic nursing care to ensure speedy recovery.

1. PATIENT WAS RELIEVED OF HEADACHE.

On the 19/08/23, the day of admission, at 12:30pm, patient complained of headache. An objective was set to relieve patient of the headache within 24 hour and the following interventions were carried out: patient was reassured of a competent nursing care to help relieve him of his headache, assessment of patient's pain was done using pain numerical rating scale, taking into consideration the onset, duration and intensity and patient rated his pain 4, all categories of health care givers were advised to minimize noise and the volume of the radio and television sets were also reduced, nursing care was organized such that it minimized interruption of patient's rest to ensure adequate bed rest, prescribe analgesics (paracetamol) and anti-hypertensive drugs (nifedipine) were administered, blood pressure chart was prepared to monitor and record blood pressure every 2 hourly, and all procedures were documented in the nurses' notes for continuity of care.

The objective was evaluated, and goal was fully met 20/08/23 at 12:30pm as patient stated relieve of his headache and also had cheerful facial expression.

2. PATIENT WAS RELIEVED OF CHEST PAIN

Interaction with patient at 12:40pm on the day of admission (19/08/2023) revealed he had chest pain. An objective was set to relieve patient of chest pain within 48 hour. Interventions carried out included; patient was reassured of a competent nursing care to help relieve him of his chest pain, assessment of patient's pain was done using pain numerical rating scale, taking into consideration the onset, duration and intensity and client rated his pain 4, gentle massages on the chest were identified to relieve patient's chest pain, patient's chest was gently massaged regularly to help relive him of the pain, nursing care was organized such that it minimized interruption of patient's rest to ensure adequate bed rest, prescribe anti-hypertensive drugs such as (lisinopril) was administered, and all procedures were documented in the nurses' notes for continuity of care.

The objective was evaluated on the 21/08/2023, and goal was fully met as patient verbalized that he nor more feels pain in the chest and also displayed a cheerful facial expression.

3. PATIENT WAS RELIEVED OF DIZZINESS.

On the 19/08/23 at 12:50am, client complained of dizziness and an objective was set to relieve patient of dizziness within 48hours and the following interventions were carried out; patient was reassured that measures will be taken to assist him regain his mobility, patient was encouraged and assisted to use Zimmer frame when moving from one area to another, comfortable bed was made for patient to promote enough rest to help stabilize his blood pressure, blood pressure chart was made to monitor patient's blood pressure every 2 hourly,

prescribed nifedipine was administered, and all procedures were documented in the nurses' notes for continuity of care.

At 12:50pm on 21/08/23, the goal was fully met as patient verbalized that he no more feels dizzy and nurse observed patient moved about the ward unassisted.

4. PATIENT WAS RELIEVED OF ANXIETY.

On the 19/08/23, the day of admission, patient was anxious. An objective was set to relieve patient of anxiety within 12 hours and the following interventions were carried out; patient was reassured that, he is in the care of competent healthcare team that would help to restore his health status, patient was encouraged to voice out fears and he stated he is uncertain if the condition was going to end his life, patient was taught of relaxation techniques like deep breathing exercises and encouraged to practice, condition was explained into details including available management to help clear his fears, patient was encouraged to listen to favorite programs on the radio to divert his mind from the anxiety, the need for hospitalization was explained to patient and his family, and all procedures were documented in the nurses' notes for continuity of care.

Objective was evaluated on 20/08/23 at 1:00pm and goal was fully met as patient verbalized that he no more feels anxious and nurse observed patient communicated freely with relatives and fellow patients without any apprehension.

5. PATIENT GAINED ADEQUATE KNOWLEDGE ON HYPERTENSION.

It was noticed through interactions with my client on the 20/08/23 at 8:00am that patient had inadequate knowledge about the condition. An objective was set to help patient gain adequate knowledge on his condition and the following interventions were put in place; patient and family were reassured that, adequate information would be provided to them for

them to gain adequate knowledge on the condition, patient and his family's level of knowledge was assessed to clarify any possible misconception about hypertension, a conducive environment was created for learning by minimizing noise and putting away all forms of destructors; thus patient's was encouraged to put his mobile phone on silence during the period of education, inter-personal relationship was created by way of sitting close to patient's bed side, this was aimed at facilitating active listening, patient and family were asked about their general health status, patient and family were educated on the causes, clinical manifestations, as well as the treatment of hypertension, and all procedures were documented in the nurses' notes for continuity of care.

The objective was evaluated on 21/08/23 at 8:00am and goal was fully met as patient was able to answer simple questions posed to him regarding the condition.

5.2 Amendment of nursing care plan.

Amendment is the process of making a change or improvement in the nursing care plan.

Amendment is made when a goal or objective is partially met or unmet. No amendments were made as all set goals were fully met within the stipulated time.

5.3 Termination of care

Termination of care is the ending of the care and relationship between the nurse, patient and relatives. Since separation can sometimes bring about anxiety, Mr. A. K. B. and his family were given a gradual psychological preparation from the day of admission to the day of discharge. They were told that the hospital was a temporal place to improve patient's condition and that he would be discharged home once he recovers.

Interaction with the patient started on the day of admission thus 19th August, 2023 through to the date of my third and last home visit, thus 10th September, 2023. Three home visits were made before care was terminated. The first home visit was on the 21st August,

2023. This was done while patient was on admission. The purpose was to verify the data I had already collected, to check if patient's environment contributed to his illness and to give health education on any health issue identified in the house.

The second home visit was carried out on 30th August, 2023. This was done after the patient had been discharged from the hospital. The purpose was to reassess the welfare of my patient and his response to treatment and to find out whether they had adhered to the education given earlier on admission and during first home visit.

The last and third home visit was made on 11th September, 2023. The purpose of the visit was to handover client to the community health nurse to ensure continuity of care, to reinforce the educations given to the patient, to see how he was doing after his review and also to terminate care. This visit was done after the review of the patient. I set off at 4:00pm and arrived at the house of my patient around 4:15pm. Everybody was doing well. Since that was my last visit, I took my time to ask for feedback on education given and highlighted on the various health educations that I had previously given. An observation was made, and all education given on the first and second visit had been put to practice. I encouraged them to continue following the instructions given on admission. They were grateful and promised to adhere to the education.

I handed over the patient to Ms. H. who is a community health nurse at the Senase CHPS, who gladly accepted and promised to visit them whenever possible. The family and patient also assured me of their cooperation with the community health nurse. I thanked them for the opportunity offered me to write a report on the care rendered to them. With this I told them that I will not visit them officially as before but assured them of friendly visits

anytime the need arises. They expressed their appreciation for the care given to them and I also thanked them for their co-operation and time given to me. I left around 5:20pm.

CHAPTER SIX

SUMMARY AND CONCLUSION

6.0 Introduction

This is the last step of the patient/family care study which entails the student's personal appreciation of the therapeutic relationship with the patient as well as the use of the nursing process.

6.1 Summary of care rendered

According to McIntosh, (2013),_summary is a short, clear description that gives the main facts of ideas about something.

Mr. A. K. B. was admitted to the male's ward of Holy Family Hospital, Berekum on 19th August, 2023 around 12:30pm. He is 76years old who was admitted with the diagnosis of hypertension. He was admitted by Dr. A. History, physical assessment and laboratory investigations were carried out to confirm his diagnosis. Patient was managed on the following drugs:

1. Tablet Lisinopril 10mg daily x 30 days
2. Tablet Nifedepine 30mg bd x 30 days
3. Tablet Methyldopa 500mg bd x 30 days
4. Tablet Paracetamol 500mg bd x 5 days

Investigations/ test ordered were:

1. Blood for full blood count
2. Blood Sample for Malaria Parasites estimation
3. Blood for lipid profile
4. Blood for fasting blood sugar

Five health problems were identified during the period of my patient's hospitalization.

These were: headache, dizziness, chest pain, anxiety and inadequate knowledge about the disease condition.

A nursing care plan for all the problems identified was drawn and implemented. Patient was discharged to go home on the 23rd August, 2023, but the nursing care continued through review and home visits and education of patient's family at home.

I made three home visits throughout my nursing care for the patient, one during the period of hospitalization, another after discharge and the last one was after review. The care given to patient at home after discharge was very good since patient and relatives adhered to the health education given to them about good personal, environmental hygiene, and balanced diet.

6.2 Conclusion and Recommendations

According to Mish (2016), conclusion is a final decision reached by reasoning. The care rendered to Mr. A. K. B. has made me gain more knowledge on the condition with regards to the cause, clinical features, medical management, nursing management and prevention.

The studies equipped me with skills on how to render total individualized care. It has also helped me improve on my interpersonal relationship with other members of the health team, the patient and family.

It is recommended that, the idea and principle behind the adoption of the nursing process which is the core approach to the writing of patient and family care study should be embraced by all nurses to ensure total patient care.

APPENDIX

Table 7: Vital signs for Mr. A. K. B.

Date	Time	Temperature (°C)	Pulse (bpm)	Respiration (cpm)	Blood Pressure (mmHg)
19/08/23 On admission	12:30pm	36.6	71	21	141/87
	2:00pm	36.8	70	20	145/90
	06:00pm	36.5	70	20	145/88
	10:00pm	36.2	68	19	140/85
20/08/23	6:00am	35.4	68	20	150/90
	10:00am	36.1	68	19	150/88
	2:00pm	36.4	69	21	148/89
	3:30pm	36.2	64	21	140/80
	06:00pm	37.2	68	19	141/80
	10:00pm	37.1	70	20	145/85
21/08/23	6:00am	36.0	76	22	150/98
	10:00am	36.4	72	19	147/89
	2:00pm	36.5	77	20	140/90
	06:00pm	35.9	70	18	148/85
	10:00pm	35.7	68	20	151/86
22/08/23	6:00am	36.5	74	24	140/90
	10:00am	36.7	68	20	140/85

Table 9: Vital signs for Mr. A. K. B. cont'd

Date	Time	Temperature (°C)	Pulse (bpm)	Respiration (cpm)	Blood Pressure (mmHg)
	2:00pm	36.5	65	18	140/75
	06:00pm	35.5	67	19	137/70
	10:00pm	36.1	65	20	130/80
23/08/23	6:00am	36.2	70	20	135/89
	11:30am	36.4	66	19	130/90
01/09/23	8:00am	36.1	76	19	127/85
On review					

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3. NURSE IN-CHARGE OF MALE MEDICAL WARD, HOLY FAMILY HOSPITAL, BEREKUM

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